

Notes: Hong & Xu (JFE, 2019)

Inferring Latent Social Networks from Stock Holdings

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1 Big picture

- **Question.** Can we infer investors' latent social networks from their stock holdings, even when the actual friendship or information network is unobserved?
- **Core idea.** A manager who has many connections in a city receives more precise private information about firms headquartered there. With short-sale constraints, greater signal precision raises the expected portfolio weight on those local stocks.
- **Statistical contribution.** The paper maps portfolio weights into latent connection counts using a portfolio-choice model and a count model of social networks.
- **Null model.** Under the Erdos–Renyi benchmark, connections are independently formed. A manager's expected connections in a city equal overall gregariousness times the city's aggregate prevalence.
- **Alternative model.** If some managers have non-i.i.d. propensities to connect to particular cities, the count distribution is overdispersed relative to Poisson.
- **Data.** The main sample uses mutual fund holdings from 1991 to 2015, merged with CRSP mutual fund data, manager demographics, stock characteristics, and headquarters-city information.
- **Main network fact.** Manager-city linkages are strongly overdispersed. Every one of the 20 major cities has an overdispersion parameter significantly above one.
- **Validation test.** The inferred relative propensity to connect to a city predicts observed university alumni links between fund managers and senior corporate officials in that city.
- **Performance test.** Fund positions in cities where a manager is inferred to have above-null connections outperform positions in cities where he is not inferred to have such connections.
- **Economic takeaway.** Portfolio holdings can reveal latent social networks. Network positions appear to be information-relevant, not merely a mechanical artifact of local bias or city size.

2 Data and measurement

2.1 Why stock holdings can reveal social networks

- Direct social-network data on investors are limited and noisy.
- Portfolio holdings are widely available and contain revealed-preference information about where investors believe they have an informational edge.
- The paper assumes that city-specific connections raise the precision of a manager’s signal about stocks headquartered in that city.
- Because funds face short-sale constraints, better signal precision increases expected positive portfolio weights.

Interpretation. The empirical strategy is indirect: it does not observe friends. It infers the latent number of city-level connections from the cross-sectional distribution of portfolio weights.

2.2 Mutual fund holdings and manager sample

- Mutual fund stock holdings come from the CDA/Spectrum Mutual Fund Common Stock Holdings database provided by Thomson Reuters.
- The sample period is 1991–2015.
- Holdings data are merged with the CRSP survivorship-bias-free mutual fund database.
- The sample focuses on active domestic equity funds managed by a single manager.
- The paper excludes international funds, bond funds, index funds, sector funds, and hedge-fund-like funds.
- The final sample has roughly 1,700 unique active domestic equity funds and about 117,000 fund-quarter observations.

Interpretation. Single-manager funds make it easier to assign portfolio choices to an identifiable decision maker with individual demographics and possible social ties.

2.3 Cities and stocks

- Stocks are assigned to metropolitan statistical areas using headquarters-city information from CRSP and Compustat.
- The paper uses the 20 largest cities because they cover about 80% of stocks held by mutual funds in the sample.
- The key city-level object is the number of latent connections manager i has in city k .
- Stock controls include size, book-to-market, momentum, and quarterly returns.

Interpretation. City-level grouping is the bridge between social networks and portfolios. If a manager’s latent contacts are concentrated in Boston, for example, the model expects more precise signals about Boston-headquartered firms.

2.4 Manager and fund covariates

- Manager demographics come from Hong and Kostovetsky-style data and internet searches where needed.
- The covariates include manager age, gender, political affiliation, undergraduate-school SAT score, and whether the manager has a graduate degree.

- Fund characteristics include total net assets, fund-family size, expense ratio, turnover, fund age, whether the fund is in a financial center, and industry concentration.
- Fund style dummies distinguish mid-cap, growth, income, and balanced funds, with micro-cap/small-cap as the omitted category.

Interpretation. The extended model uses these covariates to separate general gregariousness from non-i.i.d. city-specific connectedness.

2.5 Observable alumni networks

- RiskMetrics and Capital IQ data are used to construct alumni links between fund managers and senior corporate officials.
- A firm has an alumni connection to a university if its chairman, CEO, or CFO attended that university.
- A manager has an alumni link if he attended the same school, with the stronger measure requiring the same school, degree type, and graduation-year window.
- These alumni links are not used to estimate the latent network model. They are used to validate the inferred relative propensity to connect.

Interpretation. A good latent-network measure should predict actual observable network links. Alumni data provide an external check on the model’s inferred connections.

2.6 Return portfolios

- For each manager-city pair, the model constructs a relative propensity to connect, $\eta_{i,k}$.
- In each quarter, stock positions are split into cities with $\eta_{i,k} > 1$ and cities with $\eta_{i,k} \leq 1$.
- The paper forms value-weighted and equal-weighted portfolios from these two sets of holdings.
- The return test compares net returns and Carhart four-factor alphas of the high-RPC and low-RPC portfolios.

Interpretation. If inferred networks capture information advantages, holdings in high-RPC cities should outperform otherwise similar holdings in low-RPC cities.

3 Models and empirical specifications

3.1 Gregariousness and city prevalence

Let p_{ij} be the probability that person i is connected to person j . The expected total number of connections of person i is

$$a_i \equiv \sum_{j \neq i} p_{ij}. \quad (1)$$

For city group k , define the prevalence parameter

$$b_k \equiv \frac{\sum_{i \in S_k} a_i}{\sum_{i=1}^N a_i}. \quad (2)$$

The expected number of connections of person i in city k is

$$\mu_{i,k} \equiv \sum_{j \in S_k} p_{ij}. \quad (3)$$

The expected relative propensity to be connected to city k is

$$g_{i,k} = \frac{\mu_{i,k}}{a_i b_k}. \quad (4)$$

Interpretation. The model separates three forces: a manager's general sociability a_i , the size of the city network b_k , and the non-random city-specific propensity $g_{i,k}$.

3.2 Erdos–Renyi null model

Let $y_{i,k}$ be the number of connections manager i has in city k . Under random i.i.d. connections,

$$y_{i,k} \sim \text{Poisson}(a_i b_k), \quad (5)$$

with probability

$$\rho(y_{i,k} | a_i, b_k) = \frac{(a_i b_k)^{y_{i,k}} e^{-a_i b_k}}{y_{i,k}!}. \quad (6)$$

Interpretation. Under the null, managers can differ in overall gregariousness, and cities can differ in size, but managers do not have abnormal propensity toward specific cities.

3.3 Overdispersed network model

The alternative allows $g_{i,k}$ to vary across manager-city pairs. The model assumes

$$y_{i,k} | g_{i,k} \sim \text{Poisson}(a_i b_k g_{i,k}), \quad (7)$$

and $g_{i,k}$ is gamma distributed with mean one and variance $\omega_k - 1$. Integrating out $g_{i,k}$ yields a negative-binomial form:

$$\rho(y_{i,k} | a_i, b_k, \omega_k) = \frac{\Gamma(y_{i,k} + \zeta_{i,k})}{\Gamma(\zeta_{i,k})\Gamma(y_{i,k} + 1)} \left(\frac{1}{\omega_k}\right)^{\zeta_{i,k}} \left(\frac{\omega_k - 1}{\omega_k}\right)^{y_{i,k}}, \quad (8)$$

where

$$\zeta_{i,k} = \frac{a_i b_k}{\omega_k - 1}. \quad (9)$$

The mean is $a_i b_k$ and the variance is $\omega_k a_i b_k$.

Interpretation. The key test is whether $\omega_k = 1$. If $\omega_k > 1$, connections to city k are more concentrated than random i.i.d. link formation predicts.

3.4 Portfolio-choice model

For stock j headquartered in city k , manager i 's expected excess return is

$$E_i[\tilde{R}_{j,k} - R_f] = (\alpha + \beta X_j) + \lambda_{i,k} \xi_{i,j,k}, \quad (10)$$

where X_j contains common stock characteristics and $\xi_{i,j,k}$ is a private signal. The signal loading is

$$\lambda_{i,k} = \lambda_0 + \lambda_1 y_{i,k}, \quad \lambda_0, \lambda_1 > 0. \quad (11)$$

With risk neutrality, no short sales, and quadratic transaction costs, the optimal weight is

$$w_{i,j,k} = \frac{(\alpha + \beta X_j + (\lambda_0 + \lambda_1 y_{i,k}) \xi_{i,j,k})^+}{c}, \quad (12)$$

where $(x)^+ = \max\{x, 0\}$.

Interpretation. More connections increase signal precision. Because negative desired positions are censored by short-sale constraints, greater signal precision raises the expected positive portfolio weight.

3.5 Proposition-level implications

- **Proposition 1.** The expected portfolio weight $E(w_{i,j,k})$ is increasing in the number of connections $y_{i,k}$.
- **Proposition 2.** If the private signal is informative, stock positions in cities where the manager is more connected should outperform positions in cities where he is less connected.

Interpretation. Proposition 1 justifies inferring latent connections from portfolio weights. Proposition 2 turns the network inference into an out-of-sample return test.

3.6 Extended model with covariates

The extended model parameterizes gregariousness as

$$\delta_i = a_i \exp(\text{mgrvar}'_i \psi). \quad (13)$$

It also adds local-bias controls:

$$\exp(\kappa_1 \text{Neighbor}_{i,k} + \kappa_2 \text{Family}_{i,k}), \quad (14)$$

where $\text{Neighbor}_{i,k}$ captures holdings in city k by funds headquartered in the same city, and $\text{Family}_{i,k}$ captures holdings in city k by other funds in the same fund family.

Interpretation. These controls are designed to prevent the model from mistaking local holdings or family-level investment commonality for latent social-network structure.

3.7 Relative propensity to connect

After estimating the model, the paper constructs an RPC measure. In the baseline model,

$$\eta_{i,k} = \frac{\hat{y}_{i,k}}{\hat{a}_i \hat{b}_k}. \quad (15)$$

In the covariate model,

$$\eta_{i,k} = \frac{\hat{y}_{i,k}}{\hat{\delta}_i \exp(\hat{\kappa}_1 \text{Neighbor}_{i,k} + \hat{\kappa}_2 \text{Family}_{i,k}) \hat{b}_k}. \quad (16)$$

Interpretation. $\eta_{i,k} > 1$ means manager i appears more connected to city k than would be predicted by gregariousness, city size, and the relevant covariates.

4 Identification and empirical design

4.1 What identifies the latent network

- The model uses the geographic distribution of managers' portfolio weights across headquarters cities.
- The identifying restriction is that expected portfolio weights are monotone in latent connection counts.
- City size and manager gregariousness are separately estimated, so a high weight in New York is not automatically interpreted as a special New York connection.
- Overdispersion across managers within a city indicates non-i.i.d. link formation beyond the Erdos–Renyi benchmark.

Interpretation. The model is not simply measuring concentration. It is comparing realized city exposure with what would be expected from general sociability and city prevalence.

4.2 Why overdispersion matters

- A Poisson distribution has equal mean and variance.
- If some managers have unusual ties to a city, most managers will have few ties while a subset will have many.
- This produces variance above the mean, captured by $\omega_k > 1$.
- The paper estimates ω_k separately for each city.

Interpretation. Rejection of $\omega_k = 1$ means the network cannot be explained by random independent connection formation.

4.3 Addressing local-bias concerns

- Mutual funds may hold local stocks for reasons unrelated to social networks.
- The paper adds local-bias controls based on nearby managers' holdings and same-family holdings.
- It re-estimates the model using only New York and Boston managers and predicting their holdings in other cities.
- It also drops managers' local stock holdings from estimation.

Interpretation. The overdispersion results survive these checks, suggesting that the inferred networks are not only a local-stock-preference story.

4.4 External validation through alumni networks

- The model's RPC measure predicts observed alumni links between fund managers and corporate officials.
- The regression controls for manager demographics and city characteristics.
- A one standard deviation increase in log RPC predicts a roughly 6.5% increase in alumni links under the school-degree measure and about 4.5% under the stricter school-degree-year measure.

Interpretation. The validation test is important because it connects the inferred network measure to an independently observed social-network proxy.

4.5 Return validation

- The paper compares high-RPC and low-RPC holdings within managers.
- High-RPC holdings outperform low-RPC holdings in raw returns and four-factor alphas.
- The result is consistent with the idea that connections transmit information rather than merely creating home bias.

Interpretation. The return test gives economic content to the network inference: inferred connections predict where managers earn better performance.

4.6 Limitations

- The model infers networks; it does not observe most actual conversations or friendships.
- The interpretation relies on the maintained portfolio-choice model and short-sale constraints.
- Alumni links are only one imperfect external proxy for social ties.
- City-level networks are coarse: connections within a city can differ by firm, industry, school, and occupation.
- The model captures connections that matter for holdings, not the full social network of each manager.

Interpretation. The paper’s claim is not that it recovers every tie. It recovers a city-level latent network component that is useful for explaining holdings and returns.

5 Tables and figures

This section follows the compact visual format: adjacent figures and tables are grouped when useful, and the original images are directly cropped from the paper.

5.1 Figure 1 and Table 1: idea and sample summary

Read: Figure 1:

- The left panel is the latent social network of fund managers across cities.
- The right panel is the observed object: portfolio weights on stocks headquartered in those cities.
- The empirical task is to infer latent link counts from observed city-level holdings.

Read: Table 1:

- The mean portfolio weight on a stock in a manager-city cell is 1.33%, with a standard deviation of 1.51%.
- The baseline model predicts a mean of 3.29 contacts per manager-city cell.
- About 56% of funds are in financial centers, and the average manager age is about 49.
- The sample period is 1991–2015.

Economic message. The figure shows the inference problem. The table shows that the data are rich enough to estimate cross-manager, cross-city variation in weights and implied connections.

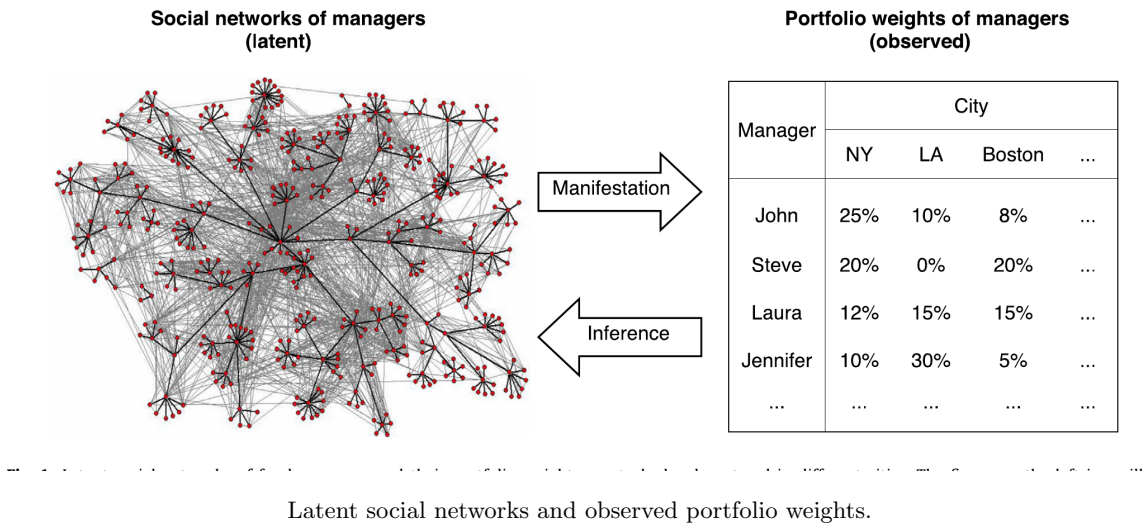


Table 1

Summary statistics of fund characteristics and fund manager demographics. This table reports the summary statistics for the funds and the fund managers in our sample. Panel A shows the summary statistics of the portfolio weights on stocks and the predicted number of connections of the funds. $w_{i,j,k}$ is a fund's portfolio weight on a stock j headquartered in a city k . $\hat{y}_{ik}$ is the predicted number of contacts a fund has in a city from our baseline model. Panel B presents the summary statistics of the characteristics of the funds. $\log TNA$, $\log FamSize$, $ExpRatio$, $Turnover$, and $FundAge$ denote, respectively, the log of the total net asset, the log of one plus the total net asset of other funds in the family, the expense ratio, the turnover ratio, and the number of years since the establishment of a fund. $FinCenter$ is a dummy variable that equals one if a fund is located in either Boston, Chicago, Los Angeles, New York, Philadelphia, or San Francisco. ICI denotes the Industry Concentration Index of a fund based on the raw industry weights of a manager's portfolio. Panel C shows the summary statistics of the demographics of the fund managers. $\log SAT$ is the log of the median SAT score (in 2005) of the undergraduate school that a fund manager attended. Adv and $Female$ are dummy variables that equal one if a fund manager holds a graduate degree and if a manager is a female, respectively. Age is the age of a fund manager, and Rep is a dummy variable that equals one if a manager is Republican-affiliated. Panel D reports the characteristics of the stocks held by the funds. $SIZE$ is the log of the market capitalization of a stock, BTM is the log of one plus the book-to-market ratio, MOM is the past 12-month return, and $Retq$ is the quarterly return. The sample period is from 1991 to 2015.

Panel A: Portfolio weights and connections					
	Mean	S.D.	Median	p10	p90
$w_{i,j,k}$ (%)	1.33	1.51	0.92	0.11	3.02
$\hat{y}_{ik}$	3.29	4.34	2.03	0.00	7.06
Panel B: Fund characteristics					
	Mean	S.D.	Median	p10	p90
$\log TNA$ (\$ million)	5.59	1.90	5.63	3.13	7.97
$\log FamSize$ (\$ million)	7.12	3.41	7.88	0.00	10.81
$ExpRatio$ (%)	1.31	1.58	1.19	0.80	1.90
$Turnover$ (%)	87.76	95.10	67.00	18.00	175.00
$FundAge$ (years)	17.50	15.41	12.99	3.75	38.75
$FinCenter$ (%)	56.14				
ICI (%)	14.85	12.43	13.73	8.38	19.64
Panel C: Manager demographics					
	Mean	S.D.	Median	p10	p90
$\log SAT$	7.11	0.14	7.12	6.94	7.27
Adv (%)	73.49				
$Female$ (%)	10.82				
Age (years)	49.47	10.37	48.00	37.00	64.00
Rep (%)	24.91				
Panel D: Stock characteristics					
	Mean	S.D.	Median	p10	p90
$SIZE$	6.22	1.79	6.11	4.02	8.57
BTM	0.45	0.26	0.41	0.16	0.76
MOM	0.13	0.61	0.04	-0.45	0.73
$Retq$ (%)	2.42	24.86	1.24	-26.09	30.31

Table 1: Summary statistics.

5.2 Tables 2–4: gregariousness, covariates, and city size

Read: Table 2:

- The mean gregariousness estimate is 63.96 in the baseline model and 72.53 with covariates.
- These estimates imply that a typical manager has roughly 70 latent friends across the 20 cities.

Read: Table 3:

- The extended model includes manager covariates, fund characteristics, fund styles, financial-center status, industry concentration, neighbor holdings, and same-family holdings.
- The estimates separate general gregariousness from city-specific network structure.

Read: Table 4:

- City prevalence parameters b_k are largest for major cities such as New York and Los Angeles.

- The city-size parameters are similar with and without covariates.

Economic message. Tables 2–4 separate “how sociable is the manager?” from “how large is the city?” before testing whether city-specific connections are non-random.

Summary statistics of estimates of coefficients on fund manager covariates. This table reports the summary statistics of the quarterly estimates of ψ , the coefficients on manager covariates $mgrvar$ in the gregariousness parameter specification $\delta_i = a_i \exp(mgrvar_i' \psi)$ in the extended model. The manager covariates include dummy variables for whether a fund is a mid-cap fund (*Medium*), a growth fund (*Growth*), an income fund (*Income*), or a balanced fund (*Balanced*). The benchmark is being a small- or micro-cap fund, i.e., the dummy variable we do not include. The covariates also include all of the fund characteristic and manager demographic variables. Specifically, *logTNA*, *logFamSize*, *ExpRatio*, *Turnover*, and *FundAge* denote, respectively, the log of the total net asset, the log of one plus the total net asset of other funds in the family, the expense ratio, the turnover ratio, and the number of years since the establishment of a fund. *FinCenter* is a dummy variable that equals one if a fund is located in a financial center. *ICI* is the Industry Concentration Index of a fund. *logSAT* is the log of the median SAT score (in 2005) of the undergraduate school that a fund manager attended. *Adv* and *Female* are dummy variables that equal one if a fund manager holds a graduate degree and if a manager is a female, respectively. *Age* is the age of a fund manager. *Rep* is a dummy variable that equals one if a manager is Republican-affiliated. *Neighbor* for a fund is the average of the portfolio weights allocated to city k of all other funds that are headquartered in the same city as the fund but not in the same family (the “know-thy-neighbor effect”). *Family* for a fund is the average of the portfolio weights allocated to city k of all other funds in the same family of the fund (the “same family effect”).

Table 2

Summary statistics of estimates of gregariousness parameters for mutual funds. The table shows the summary statistics of the estimated values of the gregariousness parameters (a_i) in the baseline model (Baseline) and $\{\delta_i = a_i \exp(mgrvar_i' \psi)\}$ in the extended model with manager covariates (With covariates), using our mutual fund data with 20 cities. We first compute the time-series average of the quarterly gregariousness estimates for each fund, then we report the summary statistics of these time-series averages.

	Mean	S.D.	Median	p10	p90
Baseline	63.96	27.23	69.58	22.76	82.76
With covariates	72.53	25.45	67.19	24.15	102.62

(a) Table 2: gregariousness.

	Mean	S.D.	Median
a_i	23.72	8.84	24.57
MEDIUM	-0.029	0.279	-0.062
BALANCED	0.078	0.174	0.033
GROWTH	0.079	0.501	0.025
INCOME	0.049	0.182	0.032
logTNA	-0.008	0.040	0.003
logFamSize	-0.008	0.029	0.000
ExpRatio	0.054	0.079	0.030
Turnover	-0.002	0.001	-0.001
FundAge	0.004	0.009	0.005
FinCenter	-0.015	0.070	0.006
ICI	0.037	0.023	0.030
SAT	0.053	0.075	0.023
Adv	0.005	0.045	0.015
Female	0.031	0.212	0.014
Age	0.005	0.011	0.010
Rep	0.002	0.118	0.016
Neighbor	0.005	0.009	0.006
Family	0.003	0.006	0.002

(b) Table 3: manager covariates.

Table 4

Summary statistics of estimates of prevalence (group size) parameters b_i for 20 cities. This table shows the summary statistics of the quarterly estimates of the relative group size parameters (b_i) for the 20 cities. The full names for the city abbreviations are as follows. NY: New York, LA: Los Angeles, Bos: Boston, SF: San Francisco, Chi: Chicago, SJ: San Jose, Dal: Dallas, Hou: Houston, Phi: Philadelphia, Was: Washington, Mia: Miami, Atl: Atlanta, Min: Minnesota, Den: Denver, SD: San Diego, Stfd: Stamford, Sea: Seattle, Phx: Phoenix, SL: St. Louis, Det: Detroit.

	Baseline			With covariates		
	Mean	S.D.	Median	Mean	S.D.	Median
NY	0.107	0.051	0.101	0.120	0.044	0.132
LA	0.057	0.034	0.057	0.065	0.025	0.074
Bos	0.054	0.023	0.053	0.067	0.025	0.070
SF	0.049	0.021	0.050	0.050	0.021	0.053
Chi	0.058	0.033	0.052	0.064	0.026	0.067
SJ	0.053	0.025	0.051	0.057	0.030	0.060
Dal	0.051	0.023	0.055	0.053	0.018	0.056
Hou	0.052	0.024	0.051	0.058	0.045	0.055
Phi	0.050	0.052	0.036	0.049	0.021	0.052
Was	0.048	0.034	0.035	0.042	0.017	0.042
Mia	0.045	0.023	0.047	0.039	0.043	0.029
Atl	0.039	0.018	0.038	0.040	0.022	0.036
Min	0.042	0.020	0.042	0.044	0.023	0.041
Den	0.043	0.034	0.039	0.035	0.025	0.030
SD	0.043	0.021	0.042	0.030	0.021	0.029
Stfd	0.042	0.027	0.044	0.049	0.048	0.029
Sea	0.046	0.036	0.041	0.036	0.031	0.026
Phx	0.047	0.038	0.037	0.037	0.038	0.023
SL	0.041	0.036	0.035	0.033	0.027	0.025
Det	0.036	0.025	0.039	0.033	0.040	0.025

(c) Table 4: city prevalence.

5.3 Table 5 and Figure 2: overdispersion and rejection of the random-network null

Read: Table 5:

- Every city has an estimated overdispersion parameter ω_k above one.
- The t-statistics against the Poisson null are significant for every city.
- New York, Los Angeles, San Jose, Miami, and San Diego have especially high overdispersion estimates.

Read: Figure 2:

- The boxplots show quarterly estimates of ω_k for each city.
- Overdispersion remains after adding covariates, although estimates are generally lower.
- The distributions are centered well above one, visually rejecting the Erdos–Renyi benchmark.

Economic message. The central result is that managers’ city connections are not random i.i.d. draws. Some managers appear to have unusually concentrated ties to specific cities.

Table 5

Summary statistics of estimates of overdispersion parameters ω_k for 20 cities. The table shows the summary statistics of the quarterly estimates of the overdispersion parameters $\{\omega_k\}$ for the 20 cities. Baseline stands for our baseline model, and With covariates stands for the extension on the baseline model where we allow for manager covariates. The t -statistics are Newey–West adjusted. They test the null hypothesis of $\omega_k = 1$ (Poisson) against the alternative of $\omega_k > 1$ (overdispersion). For an explanation of the abbreviated city names, please refer to Table 4.

	Baseline				With covariates			
	Mean	S.D.	Median	t -stat	Mean	S.D.	Median	t -stat
NY	2.301	0.945	2.223	15.01	1.973	0.680	1.963	12.19
LA	2.072	0.956	1.999	12.03	1.695	0.461	1.838	10.19
Bos	1.836	0.888	1.774	8.80	1.556	0.523	1.740	8.04
SF	1.932	0.876	1.917	10.20	1.643	0.465	1.822	9.09
Chi	1.811	0.927	1.766	10.91	1.646	0.541	1.771	9.84
SJ	2.089	0.894	2.160	10.07	1.868	0.557	2.008	8.29
Dal	1.672	0.922	1.510	12.52	1.465	0.506	1.391	8.72
Hou	1.834	0.899	1.725	12.34	1.560	0.559	1.606	9.17
Phi	1.630	0.869	1.551	13.93	1.470	0.563	1.407	12.18
Was	1.576	0.716	1.538	8.90	1.405	0.457	1.423	10.25
Mia	2.248	0.821	2.046	9.81	1.816	0.521	1.892	14.40
Atl	1.810	0.941	1.634	7.41	1.652	0.402	1.737	8.98
Min	1.774	0.795	1.677	12.78	1.687	0.492	1.623	8.49
Den	1.892	0.694	1.803	10.81	1.612	0.460	1.687	15.46
SD	2.159	0.786	2.055	12.98	1.807	0.356	1.959	19.92
Stfd	1.633	0.718	1.647	11.19	1.578	0.463	1.571	10.16
Sea	1.615	0.727	1.640	10.61	1.635	0.495	1.794	21.56
Phx	1.665	0.886	1.477	8.38	1.564	0.554	1.718	11.65
SL	1.783	0.812	1.592	20.96	1.632	0.370	1.770	10.18
Det	1.784	0.814	1.666	19.26	1.639	0.560	1.778	15.09

(a) Table 5: overdispersion estimates.

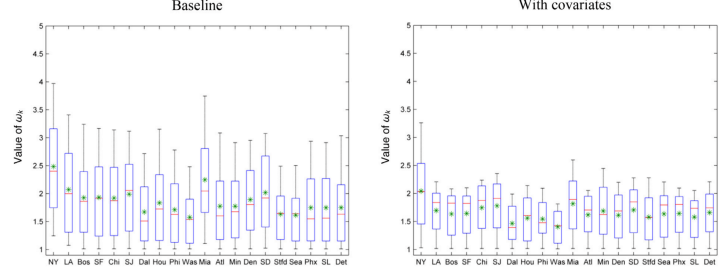


Fig. 2. Boxplot of ω_k estimates of 20 cities. The figure gives the boxplot of the quarterly estimates of the overdispersion parameters $\{\omega_k\}$ for the 20 cities from our baseline model on the left panel and from our extended model with covariates on the right panel. The * marker is the mean, the dash marker inside the box is the median, the box is the interquartile range, and the tails extend to the min and the max. For an explanation of the abbreviated city names, please refer to Table 4.

(b) Figure 2: boxplots of ω_k .

5.4 Table 6 and Tables 7–9: local bias and portfolio-choice parameters

Read: Table 6:

- The overdispersion result survives two local-bias checks.
- The first check uses New York and Boston managers and focuses on their nonlocal holdings.
- The second drops local stock holdings from estimation.
- The resulting ω_k estimates remain above one with strong t -statistics.

Read: Tables 7–9:

- Table 7 shows that fund portfolios tilt toward larger stocks and away from book-to-market and momentum after controlling for model components.
- Table 8 estimates positive signal-loading parameters: λ_0 is about 0.10 and λ_1 is about 0.02 in the baseline model.
- Table 9 shows that the RPC measure $\eta_{i,k}$ has a mean around one and meaningful dispersion.

Economic message. The robustness table says the overdispersion is not just local bias. The portfolio-parameter tables show that private signals tied to inferred connections have economically relevant portfolio-weight effects.

Table 6

Robustness checks on estimates of ω_k of 20 cities. This table shows the results of two robustness checks on the overdispersion estimates for the 20 cities. Baseline stands for our baseline model, and With covariates stands for the extended model with manager covariates. "NY&Bos funds" stands for the case where we only consider funds located in New York or Boston, and drop their picks of stocks that are headquartered in New York or Boston from the estimation. "No local response" denotes the case where managers' local holdings have been dropped from the estimation. The t -statistics are Newey–West adjusted. They test the null hypothesis of $\omega_k = 1$ (Poisson) against the alternative of $\omega_k > 1$ (overdispersion). For an explanation of the abbreviated city names, please refer to Table 4.

	Baseline				With covariates			
	NY&Bos funds		No local response		NY&Bos funds		No local response	
	Mean	t -stat	Mean	t -stat	Mean	t -stat	Mean	t -stat
NY	–	–	3.71	22.58	–	–	2.68	17.12
LA	2.42	11.15	2.41	23.71	1.75	8.67	1.56	19.29
Bos	–	–	2.55	8.93	–	–	1.75	9.72
SF	2.51	11.52	2.37	14.87	1.82	8.18	1.74	8.45
Chi	2.38	12.62	2.37	11.32	1.79	13.06	1.86	8.11
SJ	2.61	11.33	2.61	9.08	1.80	9.85	1.63	9.92
Dal	2.29	23.60	2.41	24.31	1.97	16.08	1.77	14.08
Hou	2.30	15.30	2.37	26.60	1.66	14.08	1.64	18.38
Phi	2.03	14.95	2.22	23.78	1.45	12.86	1.72	27.38
Was	2.30	11.37	2.21	10.13	1.51	12.67	1.61	9.73
Mia	2.52	10.72	2.82	12.18	1.40	16.14	1.66	21.85
Atl	2.41	7.87	2.54	7.36	2.16	10.03	2.08	10.89
Min	2.53	14.82	2.45	12.66	2.05	11.36	1.81	6.86
Den	2.78	9.52	3.00	13.39	2.16	18.64	2.12	15.30
SD	3.00	16.71	2.87	16.76	2.12	19.42	1.57	26.35
Stfd	2.15	12.33	1.97	13.14	1.61	11.77	1.60	12.71
Sea	2.17	9.12	2.18	10.09	2.03	27.31	1.83	22.60
Phx	2.40	9.15	2.25	11.91	1.96	13.89	1.64	18.78
SL	2.19	23.57	2.22	20.20	1.45	10.61	1.56	11.28
Det	2.32	22.69	2.24	24.94	1.84	16.02	1.65	22.95

Table 6: Robustness checks against local-bias explanations.

Table 7

Summary statistics of estimates of factor loadings for portfolio weights. This table reports the summary statistics of the factor loadings parameters $\beta = (\beta_{size}, \beta_{bm}, \beta_{mom})$ on stock characteristics and the constant α in the optimal portfolio choice condition of Eq. (14) where $\alpha + \beta X_i$ is the part coming from the prior and X_i are the stock characteristics. The characteristics include size (log of the market cap), book-to-market (log of one plus the book-to-market ratio), and momentum (past 12-month return), respectively.

	Baseline			With covariates		
	Mean	S.D.	Median	Mean	S.D.	Median
β_{size} (size)	0.011	0.022	0.013	0.004	0.023	–0.003
β_{bm} (book-to-market)	–0.084	0.123	–0.036	–0.078	0.185	–0.012
β_{mom} (momentum)	–0.005	0.022	–0.002	–0.009	0.030	–0.004
α (constant)	–0.233	0.218	–0.187	–0.162	0.258	–0.053

Table 8

Summary statistics of estimates of λ_0 and λ_1 for the weight $\lambda_{i,k} = \lambda_0 + \lambda_1 y_{i,k}$ on the private signal. This table reports the summary statistics of the weight parameters λ_0 and λ_1 in the optimal portfolio choice condition of Eq. (14), where $\lambda_{i,k} = \lambda_0 + \lambda_1 y_{i,k}$ is the weight on the private signal that a fund manager receives in city k and $y_{i,k}$ is the number of (latent) friends the manager has in that city.

	Baseline			With covariates		
	Mean	S.D.	Median	Mean	S.D.	Median
λ_0	0.097	0.058	0.081	0.103	0.067	0.071
λ_1	0.020	0.013	0.013	0.017	0.010	0.014

Table 9

Summary statistics of RPC measure $\eta_{i,k}$ for mutual funds. This table shows the summary statistics of our managerial RPC measure $\eta_{i,k}$ of mutual funds over all quarters. Baseline denotes our baseline model, and With covariates stands for the extended model with manager covariates. $\log(\eta_{i,k})$ is the log of one plus the value of $\eta_{i,k}$.

	Baseline	Mean	S.D.	Median
	$\eta_{i,k}$	$\log(\eta_{i,k})$		
		1.03	0.88	0.81
		0.60	0.41	0.59
With covariates	Baseline	Mean	S.D.	Median
	$\eta_{i,k}$	$\log(\eta_{i,k})$		
		1.05	0.85	0.81
		0.64	0.38	0.59

5.5 Tables 10–12: validating RPC with alumni networks

Read: Table 10:

- Harvard is the largest alumni presence for both senior corporate officials and fund managers.
- The table documents the school, degree, and cohort distribution used to construct alumni links.

Read: Table 11:

- Alumni connections are sparse, especially under the stricter school-degree-year measure.
- The school-degree measure has mean 1.38; the school-degree-year measure has mean 0.59.

Read: Table 12:

- Log RPC predicts alumni links under both alumni definitions.
- Baseline coefficients on $\log(\eta_{i,k})$ are 0.172 for school-degree and 0.133 for school-degree-year.
- With covariates, the coefficients remain positive: 0.162 and 0.110.

Economic message. The external validation is powerful: a network measure inferred only from holdings predicts actual school-based links between managers and corporate officials.

Table 10

Descriptive statistics on alumni data. This table reports the descriptive statistics of our alumni data constructed using RiskMetrics and Capital IQ databases. Panel A shows the statistics of the top five universities between 2000 and 2015 in our sample, ranked by the average number of alumni presence in firms and funds. A firm has an alumnus from a university if a senior official (Chairman, CEO, or CFO) of the firm received any degree from the university. A fund has an alumnus from a university if the fund manager received any degree from the university. Panel B shows the distribution of degrees and graduation years among company senior officials (Chairman, CEO, CFO) and fund managers in our sample between 2000 and 2015. The percentage numbers are averaged over the years.

Panel A: Universities with the highest number of alumni presence in firms and funds					
University	Ave no. of firms	Ave % of all firms	University	Ave no. of fund managers	Ave % of all fund managers
Harvard	113	10.6%	Harvard	75	12.2%
Stanford	48	4.4%	UPenn	59	9.5%
UPenn	37	3.5%	Columbia	36	5.9%
MIT	30	2.8%	NYU	34	5.7%
Columbia	25	2.4%	UChicago	32	5.2%

Panel B: Degree and graduation year distributions					
Degree	Company senior official	Fund manager	Graduation year	Company senior official	Fund manager
Undergrad	84.9%	96.8%	< 1960	10.2%	2.1%
Masters	7.5%	8.9%	1960–1969	34.1%	19.2%
MBA	16.5%	61.3%	1970–1979	40.8%	20.4%
PhD	3.7%	1.7%	1980–1989	13.2%	41.9%
Law	5.1%	1.4%	≥ 1990	1.7%	16.3%

Table 11

Summary statistics of $Alumni_{it}$ for mutual funds. This table shows the summary statistics of the number of alumni

(a) Table 10: alumni data.

Summary statistics of $Alumni_{it}$ for mutual funds. We first compute the time-series average of the quarterly $Alumni_{it}$ values (pooled across cities) for each individual fund, then we report the summary statistics of these time-series averages. School-Degree stands for the case where the alumni connection is based on a portfolio manager and a senior official of a firm (Chairman, CEO, or CFO) having attended the same school and received the same degree, and School-Degree-Year stands for the other case where the alumni connection is based on a portfolio manager and a senior official of a firm having graduated within a year from the same school and received the same degree. Since the values of $Alumni_{it}$ can be zero, $\log(Alumni_{it})$ is the log of one plus the value of $Alumni_{it}$.

Summary statistics of $Alumni_{it}$							
School-Degree	Mean	S.D.	Median	p25	p75	Min	Max
$Alumni_{it}$	1.38	3.42	0.00	0.00	1.25	0.00	32.60
$\log(Alumni_{it})$	0.52	0.82	0.00	0.00	0.79	0.00	3.51
School-Degree-Year	Mean	S.D.	Median	p25	p75	Min	Max
$Alumni_{it}$	0.59	1.58	0.00	0.00	0.50	0.00	16.60
$\log(Alumni_{it})$	0.28	0.53	0.00	0.00	0.35	0.00	2.86

(b) Table 11: alumni summary.

5.6 Table 13: RPC and portfolio returns

Read:

- The high-RPC portfolio outperforms the low-RPC portfolio in both net returns and Carhart alphas.
- In the baseline value-weighted specification, the return difference is 0.48% per quarter and the alpha difference is 0.45% per quarter.
- In the baseline equal-weighted specification, the return difference is 0.44% per quarter and the alpha difference is 0.38% per quarter.
- With covariates, the differences remain positive and statistically significant.
- The implied annual outperformance is roughly 1.5–1.8% per year.

Economic message. The return test gives the latent-network measure economic content. Managers earn better returns in cities where the model infers stronger connections.

Table 12

Using model's RPC measure to predict alumni connections. This table reports the estimation result in the regression model $\log(Alumni_{ik}) = \alpha + \gamma \log(\eta_{ik}) + x_i' \beta + v_k' \rho + \epsilon_i$ where the dependent variable is the log of one plus the number of alumni connections a fund manager i has in city k (based on our alumni connections data constructed using the Capital IQ-RiskMetrics data set). $Alumni_{ik}$ and η_{ik} are averaged across time for each fund manager i in city k . Baseline stands for our baseline model, and With covariates stands for the extended model with manager covariates. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. School-Degree stands for the case where the alumni connection is based on a portfolio manager and a senior official of a firm (Chairman, CEO, or CFO) having attended the same school and received the same degree, and School-Degree-Year stands for the other case where the alumni connection is based on a portfolio manager and a senior official of a firm having graduated within a year from the same school and received the same degree. The main explanatory variable is $\log(\eta_{ik})$, the log of one plus the RPC measure η_{ik} of manager i in city k . x_i denotes the vector of controls for fund manager demographics. For a given manager, $\logSAT$ is the log of the median SAT score (in 2005) of the undergraduate school that the manager attended, Adv and $Female$ denote whether the manager holds a graduate degree and whether the manager is a female, Age is the age of the manager, and Rep denotes whether the manager is Republican-affiliated. v_k is the vector of controls for city attributes. For a given city, $\hat{b}_k$, $\logIncome_k$, $\logPop_k$, $NonWhite_k$, $SexRatio_k$, and $OldRatio_k$ denote, respectively, the estimated value of the relative city size parameter, the log of the per capita income, the log of the population per square mile of land area, the fraction of the population that is not white, the male-to-female sex ratio, and the fraction of the population that is above 55 years old. These city attributes are obtained from the 2000 U.S. Census data. $const$ is the constant term.

	Baseline		With covariates	
	School-Degree	School-Degree-Year	School-Degree	School-Degree-Year
$\log(\eta_{ik})$	0.172*** (4.30)	0.133*** (3.13)	0.162*** (3.74)	0.110** (2.27)
$\logSAT$	0.204*** (8.13)	0.160*** (5.91)	0.207*** (8.02)	0.154*** (5.87)
Adv	0.164*** (7.60)	0.129*** (6.27)	0.166*** (7.54)	0.129*** (6.21)
$Female$	0.012 (0.77)	0.008 (1.05)	0.011 (0.71)	0.008 (1.01)
Age	0.003*** (8.40)	0.001* (1.75)	0.003*** (7.51)	0.001 (1.54)
Rep	0.064*** (4.83)	0.026** (2.28)	0.067*** (4.83)	0.027** (2.39)
$\log(\hat{b}_k)$	0.838*** (9.13)	0.441*** (7.94)	0.470*** (10.74)	0.242*** (7.74)
$\logIncome_k$	0.706*** (4.70)	0.444*** (5.17)	0.338** (2.29)	0.257** (2.53)
$\logPop_k$	0.102** (2.66)	0.077*** (3.48)	0.083* (1.98)	0.067** (2.50)
$NonWhite_k$	0.137 (0.52)	-0.050 (-0.35)	0.098 (0.58)	-0.065 (-0.70)
$SexRatio_k$	-1.353*** (-5.12)	-0.871*** (-5.12)	-0.489 (-0.88)	-0.444 (-1.35)
$OldRatio_k$	-1.167*** (-4.96)	-0.552*** (-3.14)	-0.100 (-0.14)	0.361 (0.93)
$const$	-3.248*** (-4.49)	-1.913*** (-4.62)	-2.171*** (-4.66)	-1.296*** (-4.13)

Table 13

Returns for portfolios constructed using RPC measure. This table reports the returns of the portfolios constructed based on our RPC measure η_{ik} at the quarterly level. In each quarter, the $\eta_{ik} > 1$ portfolio of a manager i is the portfolio based on his stock holdings from those cities where he has $\eta_{ik} > 1$, and the $\eta_{ik} \leq 1$ portfolio is the portfolio based on his stock holdings from those cities where he has $\eta_{ik} \leq 1$. We then construct the value-weighted or equal-weighted $\eta_{ik} > 1$ portfolio across all managers (the $\eta_{ik} > 1$ portfolio) by value-weighting or equal-weighting all fund managers' $\eta_{ik} > 1$ portfolios, and similarly the $\eta_{ik} \leq 1$ portfolio. The $\eta_{ik} > 1$ portfolio and the $\eta_{ik} \leq 1$ portfolio are held for one quarter, and then the two portfolios are rebalanced based on the η_{ik} 's. The quarterly net returns of the two portfolios (net of risk-free rate), their return differences Diff (return of $\eta_{ik} > 1$ portfolio minus $\eta_{ik} \leq 1$ portfolio), and their associated Carhart four-factor alphas are shown, together with their t -statistics (Newey–West adjusted).

Panel A: Baseline				
Portfolio	Value-weighted		Carhart alpha	
	Mean (%)	t-stat	Mean (%)	t-stat
$\eta_{ik} > 1$	1.46	2.63	1.03	2.20
$\eta_{ik} \leq 1$	0.98	1.67	0.58	1.01
Diff	0.48	2.07	0.45	2.10
Panel B: With covariates				
Portfolio	Value-weighted		Carhart alpha	
	Mean (%)	t-stat	Mean (%)	t-stat
$\eta_{ik} > 1$	1.37	3.49	0.77	2.19
$\eta_{ik} \leq 1$	0.93	1.99	0.39	0.98
Diff	0.44	2.21	0.38	2.29
Panel C: With covariates				
Portfolio	Value-weighted		Carhart alpha	
	Mean (%)	t-stat	Mean (%)	t-stat
$\eta_{ik} > 1$	1.42	2.53	1.02	2.07
$\eta_{ik} \leq 1$	0.99	1.71	0.60	1.23
Diff	0.43	2.14	0.42	2.24
Panel D: With covariates				
Portfolio	Value-weighted		Carhart alpha	
	Mean (%)	t-stat	Mean (%)	t-stat
$\eta_{ik} > 1$	1.34	3.27	0.76	2.10
$\eta_{ik} \leq 1$	0.95	1.95	0.40	1.00
Diff	0.39	2.15	0.36	2.37

6 Short summary

- Hong and Xu develop a method to infer latent social networks from mutual fund holdings.
- The model starts from a random-network benchmark in which expected city connections equal manager gregariousness times city prevalence.
- It allows non-i.i.d. city-specific propensities to connect, producing an overdispersed negative-binomial count model.
- Portfolio weights identify latent connection counts because connection-driven signal precision raises expected positive weights under short-sale constraints.
- The empirical sample uses active single-manager domestic equity funds from 1991 to 2015.
- All 20 major cities have overdispersion estimates significantly above one.
- The overdispersion remains after controlling for manager covariates, neighbor holdings, same-family holdings, and local-bias checks.
- The inferred relative propensity to connect predicts observed university alumni networks between fund managers and corporate officials.
- High-RPC positions outperform low-RPC positions by about 0.4–0.5% per quarter.
- The paper shows that holdings data can be used to infer latent social networks that are both observable-network-relevant and return-relevant.

7 Conclusion

7.1 Core conclusion

Hong and Xu show that latent social networks can be inferred from stock holdings by combining a portfolio-choice model with an overdispersed count model. Managers' city-level portfolio weights imply non-random social connections across cities, and these inferred connections are validated by alumni networks and by portfolio performance.

7.2 Economic intuition

The intuition is that a social connection is valuable if it improves the precision of information about firms in a city. A manager with many connections in a city receives more precise private signals. Because funds cannot short freely, greater signal precision increases expected long portfolio weights. The cross-sectional dispersion in these weights can therefore reveal who is unusually connected to which cities.

7.3 Takeaway

The paper provides a practical way to recover otherwise unobserved social networks from investment behavior. Its broader lesson is that portfolio holdings are not merely outcomes of preferences and risk exposures; they also contain information about the hidden network structure through which investors acquire private information.